

Stage 1 Modeling

Tree morphology measurement
 $\mathcal{G} = (\mathcal{B}, \mathcal{E})$



Branch material tests
→ priors $p(\theta)$



Fruit distribution model
(linear arc-length density)



Assemble FE operators
 $\mathbf{M}, \mathbf{K}(\theta), \mathbf{C}(\alpha, \beta)$



Local nonlinear correction
 $k_3 \Delta u^3 + c_2 |\Delta v| \Delta v$



**Prior-constrained
FE model**



Stage 2 Calibration

Hammer test
(force + accel. signals)



Butterworth LP filter
 $f_c = 50$ Hz, zero-phase



H_1 FRF estimate
 $H_1 = G_{af}/G_{ff}$



Peak picking
 $\hat{f}_r, |\hat{H}(\omega_i)|$



MCMC calibration (emcee)
 $\theta \in \mathbb{R}^8$



Posterior
 $p(\theta | \hat{f}, \hat{H})$



Stage 3 Validation

Fixed-frequency
excitation test



Forward solve at $\hat{\theta}$
(posterior median)



Posterior predictive FRF
(NRMSE, peak error)



90% CI coverage check
vs. independent data



Linear vs nonlinear FRF
 $\Delta f/f_{iin}$, peak shape



**Validated
calibrated model**



Stage 4 Recommendation

(p, f, A) sweep
 $\mathcal{P}_0 \times \mathcal{F} \times \mathcal{A}$



Multi-clamp Pareto
 C_{cov} vs σ_{tr}



Knee selection
 $\min \|(\tilde{y}_1, \tilde{y}_2)\|_2$



Posterior + Sobol
90% CI, $S_{T,i}$



Operating-parameter sequence
 $\{(f_k, A_k)\}_{k=1}^N, C_{cov}^{cum} \uparrow$



Staged recommendation
 $(p^\dagger, \{(f_k, A_k)\}_{k=1}^N) \pm \text{CI}$